

Title

Effects of structured exercise on fibroblast growth factor 21 and adiponectin dynamics in older women with metabolic syndrome: study protocol

Abstract

Metabolic syndrome impairs glucose homeostasis, lipid metabolism, and cardiovascular health in aging populations. The FGF21-adiponectin signaling axis regulates systemic metabolic adaptation, but responsiveness to exercise training may differ based on baseline metabolic status. This protocol describes a randomized trial evaluating the effects of 10 weeks of supervised combined exercise on circulating FGF21, adiponectin concentrations, and their ratio as primary biomarkers in 60 older women with or without metabolic syndrome. The exercise intervention comprises resistance training and elastic-band conditioning three times weekly. Control participants maintain usual activity patterns. Blood samples, body composition via dual-energy X-ray absorptiometry, physical fitness assessments, and metabolic parameters are measured at baseline and post-intervention. Two-way repeated-measures analysis will assess exercise effects stratified by metabolic status. This investigation aims to clarify mechanisms of exercise-induced metabolic improvement and identify biomarker-based responder phenotypes.